



RIVERDALE MIDDLE SCHOOL

UNESCO HACKATHON PROPOSAL · CRITICAL THINKING & AI EDUCATION / MEDIA & INFORMATION LITERACY

I Thought I Knew You

A critical-thinking and AI-literacy game disguised as a four-night group-chat mystery — teenagers learn to reason about a deepfake video, a misdated photo, a cloned social account, and a cloned voice, using a friendship they care about as the reason to bother.

TEAM

— add your names here —

TRACK

— add hackathon track —

FORMAT

Browser game · client-side AI

AUDIENCE

Secondary school, ages 13+

What's in this document

A game about one thing: teaching teenagers to reason carefully about AI-manipulated media, using a friendship they care about as the reason to bother.

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Critical thinking hasn't caught up with generative AI

Generative AI has made convincing fabrication cheap: a voice cloned from a few seconds of audio, a face swapped into a photo, a caption that reframes a real image into a lie. The reasoning skills needed to catch this are teachable — but almost nobody is teaching them where it counts. For teenagers, manipulated media doesn't arrive as an abstract news story to fact-check calmly; it arrives as a video forwarded into a group chat by someone they trust, about someone else they trust, with everyone already reacting before anyone has thought to check anything.

STRIPPED CONTEXT

Real media, false frame

A genuine clip or photo, cut or re-dated so it means something it didn't mean.

SPOOFED IDENTITY

Fake accounts, cloned voices

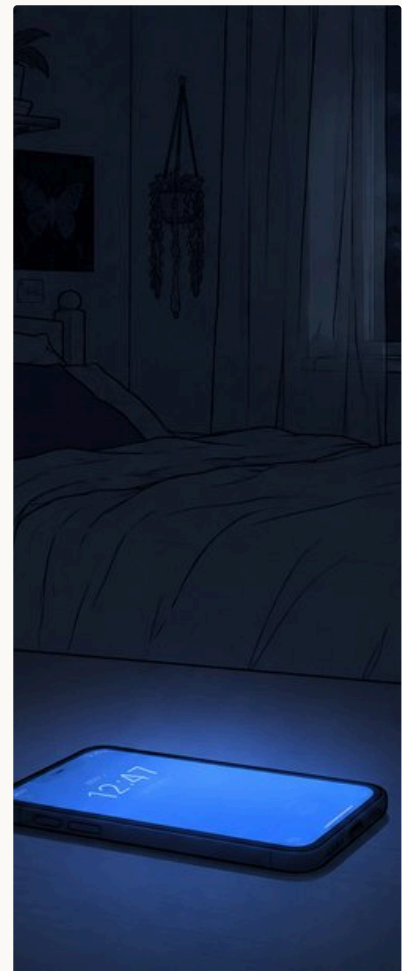
AI-generated images and synthetic audio built to pass as someone real.

SOCIAL PRESSURE

Reacting before checking

Group chats reward speed of reaction, not accuracy — verification feels slow and socially costly.

None of these three failure modes are solved by teaching young people to "spot the fake" in isolation — the video in this scenario, for instance, isn't fake at all. What's missing isn't detection skill in the abstract; it's the **habit** of pausing, tracing a claim back to its source, and asking who benefits from you reacting fast. That is a behavioural gap, not a knowledge gap — which is why this project is built as a game about consequences, not a quiz about facts.



Real techniques, modelled directly

Every piece of "evidence" in the game is built from a manipulation technique already circulating online today — nothing in the scenario is speculative or futuristic.

Selective re-cutting of real video

Recycled photo, new date/caption

AI face-swapped impostor account

Cloned voice message

False "AI-authentic" confidence badge

The framework: DW Akademie's MIL Index

The project's primary goal is critical thinking and AI literacy — the friendship story is the delivery mechanism, not the message. That goal is built on a named competency framework for what "critical thinking about media" actually consists of: this hackathon's own reference model for media and information literacy.

1 • The MIL Index model



- Access** — using media technology to reach a message in the first place, and knowing where to look for information.
- Analyze** — interpreting and critically evaluating a media message against what you already know.
- Reflect** — critically examining which sources were actually used, and what impact that choice had.
- Create** — composing and sharing a message of your own to express an idea or opinion.
- Act** — putting the skill into practice, for your own benefit and your community's.

Diagram and definitions adapted from DW Akademie's Media and Information Literacy Index, the "MIL Heroes" model (DW Akademie, 2020) — the reference framework behind this hackathon's "Heroes and Villains in the Age of AI" brief. Full pillar-by-pillar mapping to specific game mechanics appears in section 11, "Alignment with the MIL Index." Full citation on page 6.

Most "media literacy" games only exercise Analyze — spot the fake. This one was designed to exercise all five pillars in a single playthrough, not detection in isolation.

Why this exact model, and not a generic one

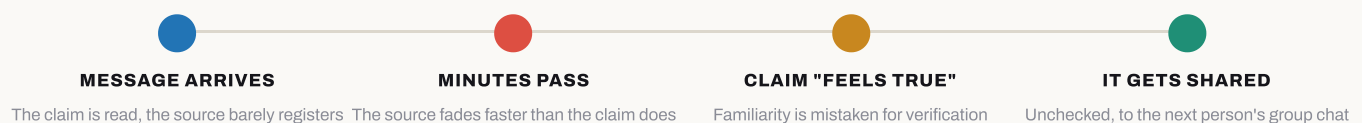
This hackathon's own brief is already anchored on the MIL Index, so building against it isn't an add-on for this document — it's the same yardstick judges are already using. Treating "critical thinking" as five distinct, separately-trainable competencies, rather than one vague instinct, is also what makes the game's design falsifiable: each pillar below is tied to a specific, checkable mechanic (section 11), not a vibe.

Why people get fooled, and how to rehearse resistance

Knowing the MIL pillars explains *what* to train. These two ideas, both with a research track record in misinformation studies, explain *why people fail* and *how rehearsal fixes it*.

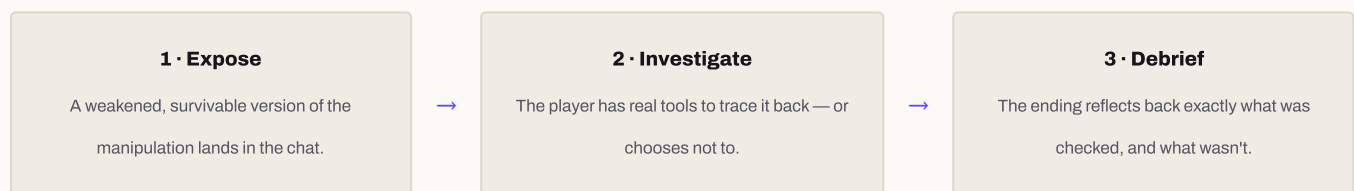
2 · The psychology of misinformation

People are rarely fooled because they're gullible. Memory research on **source monitoring** — the process of attributing a memory or belief to where it actually came from — shows that people remember the content of a claim far better than its origin, so a plausible claim "feels true" once the source is forgotten (Johnson & Raye, 1981). Later work found this same mechanism drives the "misinformation effect," where a false detail gets misattributed to a real memory (Lindsay & Johnson, 1984). Critical thinking, in this framing, isn't a personality trait some people have and others lack; it's a specific, teachable habit of re-attaching claims to their source before reacting — which is exactly what the game's investigation actions drill.

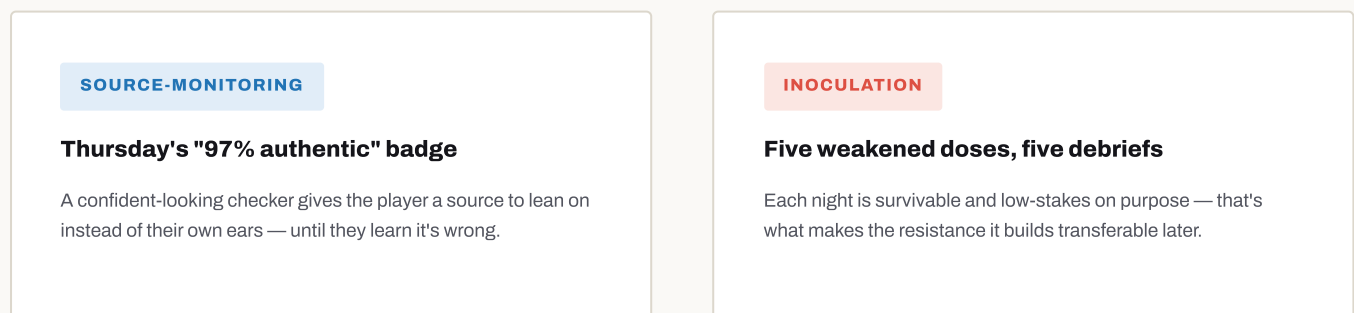


3 · Prebunking / inoculation theory

Resistance to manipulation transfers better when it's rehearsed in a low-stakes setting before the real exposure — a weakened "dose" of the manipulation technique, with the trick explained immediately afterwards, the same logic behind medical inoculation. Controlled studies of this approach, including the browser game *Bad News*, found that pre-exposing people to weakened misinformation techniques measurably improved their ability to spot manipulation afterwards, across cultures (Roozenbeek & van der Linden, 2019; Basol et al., 2021). Each of the five nights follows the same structure, repeated until it becomes a reflex rather than a rule to remember:



Both ideas, in one game moment



Why it's a story, and why it's a game

The MIL pillars and the psychology of belief explain what to train and why rehearsal works. These two ideas explain why that rehearsal is wrapped in a first-person story instead of a lecture or a checklist.

4 • Narrative transportation & perspective-taking

Audiences who become absorbed, or "transported," into a story pay closer attention, form vivid mental imagery, and are measurably more likely to adopt the beliefs and attitudes the story implies than audiences given the same information directly (Green & Brock, 2000). This is why the vehicle is a friendship, not a worksheet. The fifth night pushes this further and deliberately reverses the camera — the player becomes the one being falsely accused — turning "I can judge Nicole's evidence" into "I understand exactly how this gets done to someone," a stronger, stickier form of the same critical-thinking lesson.

5 • Experiential / game-based learning

Kolb's experiential learning cycle holds that durable learning comes from a loop of concrete experience, reflection, and active experimentation, not from being told a conclusion (Kolb, 1984). Games are a natural fit for that loop: Gee's study of learning in well-designed video games found that they routinely embed exactly this structure — act, get real feedback, adjust — better than most classroom formats manage (Gee, 2007). The game never states a rule and asks the player to recall it; it withholds the answer, lets a real (in-fiction) consequence follow from the player's actual choice, and only then shows what was true.

Teaching "how to spot a deepfake" is a knowledge problem. Teaching someone to *feel the urge to check* before they forward something is a behavioural one — and behaviour is what rehearsed, consequence-driven game mechanics are good at shaping.

IDEA	WHERE IT ACTUALLY SHOWS UP IN THE GAME
Narrative transportation	The whole friendship framing, and night five's camera reversal onto the player.
Experiential learning	No stated rules — only actions, consequences, and a debrief that shows what was true.

References

DW Akademie. (2020). *Media and Information Literacy Index — "MIL Heroes."* Deutsche Welle Akademie.

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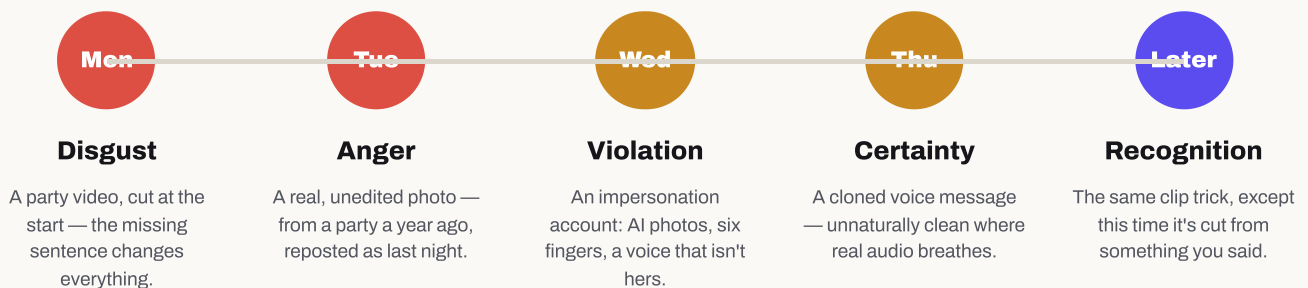
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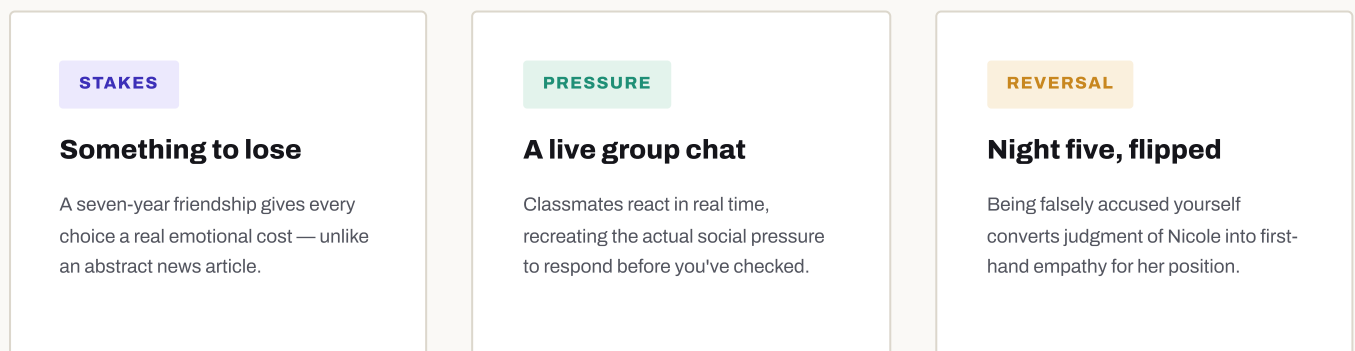
Five nights, five reasoning exercises, one friendship

Structurally, this is five separate exercises in evaluating AI-manipulated media, each with a different manipulation technique and a different detectable tell. Narratively, it's one continuous story: the player is one of 28 classmates in a group chat, and over four evenings the chat receives one piece of damning "evidence" against Nicole, the player's best friend of seven years — each staged as something that just landed on your phone, not as a lesson. A closing fifth scenario turns the same trick on the player themselves.



Every "tell" is real and checkable — a full-length video that recontextualises the clip, a dated post that recontextualises the photo, a visual or stylistic artefact in the fake account, a room-tone difference in the voice message. Nothing requires the player to be told the answer; it requires them to look. The title pays off literally at the end of night one, when a classmate posts: *"wow. i thought i knew her."*

Why a friendship, not a headline



Every one of these five nights resolves the same way: the truth was checkable the whole time, and the player either checked or didn't. Nothing in the story is unknowable — that's a deliberate design constraint, not a coincidence.

Investigate. React. Or say nothing at all.

Each evening is a self-contained reasoning task dressed as a phone screen. The player has limited time to spend on three kinds of moves: **investigate** (reverse-image search, ask a witness, compare accounts, replay audio slowed down) — the actual critical-thinking work — **react in the group** (forward, pile on, defend, stay silent), or **message Nicole privately**. Every choice moves a friendship score, the group's opinion of the player, and which of several endings is reached, so reasoning well and reasoning fast are put in direct, felt tension.

NIGHT	EVIDENCE	WHAT'S ACTUALLY TRUE	THE DETECTABLE TELL
Monday	A party video	Real footage, real party — but cropped before the sentence that explains it.	Ask for the full clip.
Tuesday	A party photo	Completely real and unedited — just from a party a year earlier.	Check the original post date.
Wednesday	A second account	Not her — an AI-generated impersonation. Her real, silent account is still live.	Compare the two accounts.
Thursday	A voice message	An AI voice clone; the "97% authentic" checker badge is simply wrong.	Listen for room tone twice.

The chat, in character

Every classmate holds a fixed emotional stance, delivered live through a small AI language model running in the player's own browser (see *Technology*, next). No two conversations play identically, but no character ever contradicts the true story.



Nicole

The accused. Direct, casual, imperfect — worried, but not performing it.



Hanna

Certain, dismissive, quick to pile on. Opposes Nicole from night one.



Lea

Follows Hanna's read of the room, rarely first to speak.



Mia

The skeptic. Questions the evidence if the player gives her a reason to.



Benito

Evasive. Says little — but he knows more than he says.

The three moves, and what each one trains

ACCESS + ANALYZE Investigate Costs real time — the reasoning work.	SOCIAL PRESSURE React in the group Instant, public, and free — tempting.	CREATE + ACT Message Nicole Private and direct, away from pressure.
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Every night's actual reasoning task

Each investigation action below is a real, selectable move from the game — the exact label the player sees, and what checking it actually reveals. Some are genuine progress; a few are deliberate dead ends, because knowing when verification effort isn't paying off is part of the skill.

 Monday — the video DISGUST · REAL FOOTAGE, CUT BEFORE THE START

- **"Watch it again, slowly"** — the clip starts mid-word, proving something was cut.
- **"Ask Mia, she was at that party"** — she confirms Nicole was quoting someone else.
- **"Ask who filmed it"** — Hanna admits she doesn't know, just received it.
- **"Reverse search the video"** — dead end, no matches (costs time, teaches nothing).

 Tuesday — the photo ANGER · GENUINELY REAL, JUST FROM A YEAR EARLIER

- **"Reverse search the photo"** — the same file was uploaded back in July.
- **"Look at the background"** — a half-hidden banner reads "LAURA'S BIRTHDAY."
- **"Look at the photo properly"** — bare arms outdoors at night; wrong for November.
- **"Ask Nicole about the photo"** — she hasn't left her room in two days: an alibi.

 Wednesday — the second account VIOLATION · AI IMPERSONATION

- **"Put the two accounts side by side"** — real: 52 posts since 2022; fake: 4 posts, all this week.
- **"Compare the writing to her real captions"** — real Nicole never uses full stops or capitals; the fake does both.
- **"Look at the followers"** — the fake's 42 followers all followed this week.
- **"Ask Nicole if the account is hers"** — she denies it directly.

 Thursday — the voice message CERTAINTY · CLONED VOICE, "97% AUTHENTIC" CHECKER IS WRONG

- **"Listen closely, twice"** — no room tone, no breathing, identically-timed pauses.
- **"Ask where she was tonight"** — Mia confirms Nicole was with family all night.
- **"Compare it to her old voice notes"** — real ones trail off and laugh; this one never does.
- **"Run the AI checker yourself"** — returns "97% AUTHENTIC" — a deliberate false-confidence trap.

 Later — the clip of you RECOGNITION · YOUR OWN WORDS, SPLICED

- **"Compare it to the original"** — matches your real Sunday voice note; same words, different order.
- **"Listen again, slowly"** — a phrase cuts off mid-word instead of trailing off naturally.
- **"Ask when it was recorded"** — Nicole herself notices the timestamp doesn't add up.

Time is the real currency

Every action above costs limited in-game time, so "check everything" is never free — reasoning well under a time constraint is the actual skill being trained, not reasoning well with unlimited time. The debrief on the final night (see *Educational Impact*) measures exactly where that time went.

- Investigate — costs time
- React in the group — instant, public
- Message Nicole — private, direct
- Report an account — reversible consequence

Wednesday: which account is real?

This is the actual in-game comparison screen from the "second account" scenario — the clearest single example of what the player is asked to practise: not "trust your gut," but look for the specific, findable tell.

Nicole's real account4 YEARS OLD

← Profile



Nicole

Aenean interdum risus nulla in purus convallis volutpat consectetur mi rhoncus eu lacinia est amet

40

120

49

Postingan

Follower

Following



Nicole 1

Jakarta, Indonesia

Pharetra, risus erat nibh facilisis imperdiet in ut mauris non urna gravida nulla quam in

13.868

467

1.675

The copycat accountDAYS OLD

← Profile



Nicole Fake

Aenean interdum risus nulla in purus convallis volutpat consectetur mi rhoncus eu lacinia est amet

2

30

2

Postingan

Follower

Following



Nicole Fake

Jakarta, Indonesia

Pharetra, risus erat nibh facilisis imperdiet in ut mauris non urna gravida nulla quam in

13.868

467

1.675

— Long, ordinary posting history

— Photos and captions match her established style

— Has gone quiet — she isn't posting to defend herself

— Same name and photos, wrong writing voice

— AI-image artefacts on close inspection

— Posting fast, right when it's most damaging

The two screens are visually similar on purpose — this is the point. Judges and players alike are meant to have to actually compare them side by side, the same way the in-game "compare accounts" action works, rather than intuit which is fake at a glance.

I THOUGHT I KNEW YOU — UNESCO HACKATHON PROPOSAL

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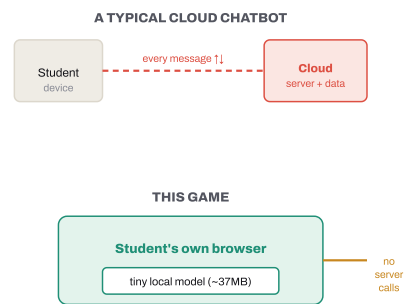
An AI that runs on the player's device, not a server

The in-character chat is powered by a small, quantised language model (Falcon-H1-Tiny-90M, ~37 MB) run entirely client-side in the browser via WebAssembly, with optional WebGPU acceleration. Every reply is grounded against a short, per-day fact sheet the player never sees, so improvised dialogue can never contradict the "true" version of events. This is also, quietly, AI education in itself: the player is talking to a real, working generative model the whole time — a concrete, hands-on referent for "what an AI actually is" underneath every abstract warning about deepfakes.

Why that matters for schools

- **No account, no server, no data collection** from minors — nothing typed by a student leaves their own device.
- **Works after one load**, even on slow or filtered school networks, since there is no ongoing API call per message.
- **No build step.** Plain static files — deployable from a USB stick, a school server, or free static hosting, with zero infrastructure cost.
- **Content lives in editable JSON**, separate from code — a teacher, translator, or student team can extend or localise scenarios without touching a line of logic.

Typical AI chat vs. this game



Model card

Model	Falcon-H1-Tiny-90M-Instruct, GGUF Q4_K_M quantisation
Footprint	~37 MB, downloaded once and cached in the browser
Runtime	wllama (llama.cpp compiled to WebAssembly), optional WebGPU acceleration, CPU fallback
Grounding	A short per-day fact sheet constrains every reply so improvisation can't contradict the true story
Guardrails	Replies breaking character, breaking the fourth wall, or sounding like an assistant are rejected and retried

The lesson is the ending, not a warning label

There is no "did you spot the deepfake?" quiz. Instead, the game silently tracks what the player actually did all week, and hands it back to them, unfiltered, at the end.

What gets measured

A real excerpt from the in-game end-of-week ledger:

Forwarded	— things
Reacted	— times
Reported her real account	yes / no
Reported the fake account	yes / no
Time spent checking vs. reacting	— min / — min

What gets said back to you

Real debrief lines, selected by what actually happened in the playthrough:

- "You never looked for the original of anything. Originals existed."
- "You never compared the account to her real one. It was still up."
- "You checked all five. That's rarer than it should be."

Because the debrief is generated from tracked behaviour rather than a fixed script, no two players are told the same thing — the feedback is as specific as the choices that produced it. This turns the ending into a private, individual mirror rather than a group lecture, which is precisely the self-assessment moment MIL frameworks call for and which a classroom rarely has time to deliver one-to-one.

The same night, three different closing lines

VERIFIED & SUPPORTIVE

"She checked it. She learned that from you."

RIGHT, BUT LATE

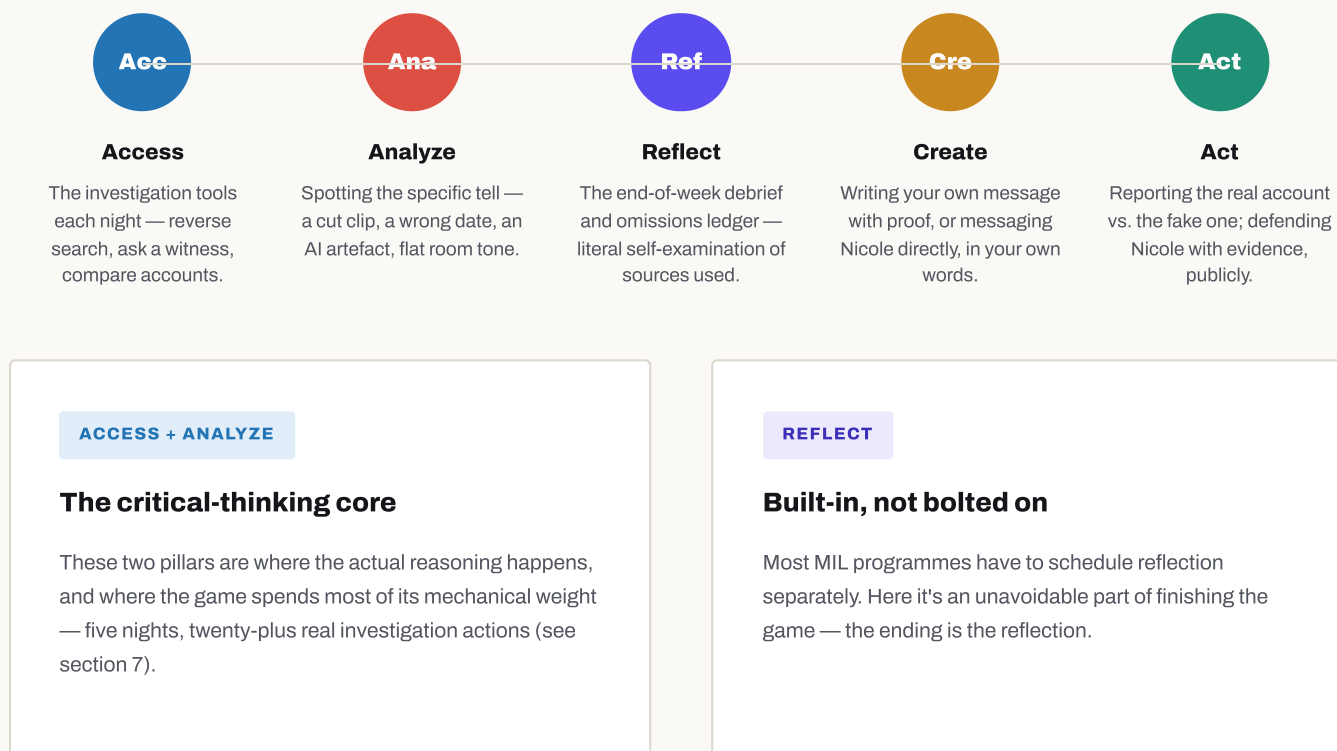
"You knew by Tuesday. It changed nothing."

NEVER CHECKED

"I thought I knew you, {name}."

Every pillar, one concrete mechanic each

Section 2 introduced DW Akademie's MIL Index — the five-pillar model this hackathon's own brief is built around. Rather than claim generic alignment, here is exactly which game mechanic exercises each pillar, in the order a player actually encounters them: detect first, reflect after, then create and act.



In one sentence: this is a MIL Index exercise disguised as a story about a friendship — every pillar exercised, in the same order, every single playthrough.

Equity note: none of the above requires a server, an account, or reliable connectivity after first load — which matters for Access specifically, since a pillar about "knowing where to find information" is undermined if the tool itself is hard to reach in an under-resourced classroom. This describes thematic alignment with DW Akademie's published MIL Index model; it is not a claim of formal review, partnership, or endorsement.

From hackathon build to classroom tool

1 Facilitator guide

A short companion document for teachers: discussion prompts per night, and how to read the anonymised end-of-week ledger as a class discussion starter rather than a grade.

2 Localisation

All narrative content already lives in separate JSON files, decoupled from game logic — translation into other languages is a content task, not an engineering one.

3 Accessibility pass

Screen-reader support for the chat interface, captions for the voice-message evidence, and colour-contrast review of the investigation UI.

4 More scenarios

Additional evidence types (manipulated group photos, forwarded-chain misinformation) reusing the same investigate / react / debrief structure.

5 Aggregate, opt-in classroom view

An anonymised summary of common omissions across a class set, to focus a follow-up lesson on the habits a specific group actually struggled with.

Static hosting · GitHub Pages

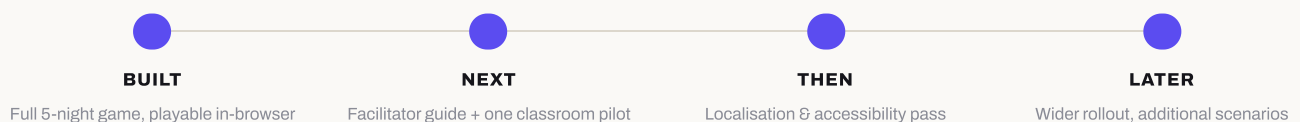
No build step

Content in JSON

Client-side LLM

No student data collected

Where this stands



Who built this

NAME	ROLE	CONTACT
— add name —	— add role —	— add contact —
— add name —	— add role —	— add contact —
— add name —	— add role —	— add contact —

Team photo, logo, or additional credits — placeholder

Try it / find it

Live game

— add deployment link —

Source

— add repository link —

"I thought I knew you."

— the line every ending in the game earns differently, depending on what the player actually checked.

Thank you for reading. Questions welcome before, during, or after the pitch.